

Arthur Torres

AI ENGINEER

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SUMMARY

I architect where AI belongs in an operational pipeline: what becomes an agent, what stays deterministic code, and what should not be built at all. At Banco Inter's Global Operations team, that shipped as LLM agents in production under audit constraints, plus **nine automations saving 1.4 FTE in 7 months**. Also ships end-to-end ML pipelines, from 37M-record telemetry modeling to LLM document intelligence.

EXPERIENCE

Banco Inter

Jan 2026 to Present

AI & Automation Intern · Global Operations · Belo Horizonte · On-site

- **LLM agents in production:** built in code on an internal LiteLLM gateway, owning system prompts, tools and MCP integrations, for governance workflows in a regulated bank
- **Onboarding Decision Engine:** evaluates onboarding payloads and returns structured decisions to analysts via API, eliminating manual review (*0.4 FTE saved*)
- **Account Alert Automation:** end-to-end email pipeline for accounts requiring financial attention, triggered with zero manual intervention (*0.3 FTE saved*)
- Three more tools: LLM document analysis agent, address validation pipeline (FastAPI + web scraping), transaction integrity validator. Saving 0.3 FTE combined, all deployed as microservices on Kubernetes

Python · SQL · FastAPI · LLM Agents · MCP · LiteLLM · Selenium · Pandas · Docker · Kubernetes · GitLab CI/CD

CEFET-MG

Aug 2026 to Present

Undergraduate Researcher · LLM Models for Market Prediction and Risk · Belo Horizonte · Hybrid

- Hybrid models combining time series analysis, portfolio optimization and LLMs to forecast market trends and manage financial risk
- Integrating numerical and textual signals: historical price series, financial news and sentiment analysis, aligned on the same timeline
- Evaluating the combined method against each approach on its own, rather than reporting a single number in isolation

Python · LLMs · Time Series · FinBERT · Portfolio Optimization

Akyou Soluções

Apr to Dec 2025

Backend Engineer · AI & Accessibility · Remote

- Built the Python + OpenAI backend and LLM pipelines restructuring unstructured PDFs into accessible output
- Took the product from scratch to a working prototype, owning it end to end: TypeScript frontend built from the ground up, backend deployed on AWS Elastic Beanstalk behind a load balancer

Python · TypeScript · OpenAI API · LLM Pipelines · AWS Elastic Beanstalk

PROJECTS

Financial Intelligence Dashboard github.com/tutorres/Financial-Intelligence-Dashboard · [Live demo](#)

Python · DuckDB · PyTorch · Groq API · Streamlit · Pandas · Plotly

Medallion pipeline (bronze/silver/gold), LSTM trend classification, Text-to-SQL LLM interface, market chat.

Vale Desenvolver 2026 · Mining Fleet Telemetry github.com/tutorres/Vale_Desenvolver · [Case study](#)

Python · XGBoost · LightGBM · SHAP · DuckDB · pytest

Predictive pipeline over 37.1M telemetry records from an iron ore mine in Itabira, in an open challenge run by Vale S.A., anticipating critical "Don't Go" failures: F1 0.186 at 4h vs 0.038 for the best heuristic (4.9x). Temporal validation, SHAP, 93 tests (TDD).

SKILLS

AI / ML LLM Agents · MCP · OpenAI / Groq API · PyTorch · LSTM · XGBoost / LightGBM · SHAP

CORE Python (advanced) · SQL (advanced) · FastAPI · Selenium · pytest · Git

DATA

DuckDB · Pandas · Plotly · Streamlit · OpenPyXL

INFRA

Docker · Kubernetes · GitLab CI/CD · AWS

PRACTICES

TDD · Spec-Driven Development · Agentic coding workflows

EDUCATION

CEFET-MG · B.Eng. Computer Engineering

2024 to 2030 (est.) · 5th semester

CERTIFICATIONS & LANGUAGES

Programa Desenvolver 2026 · Data Analysis Vale S.A. · challenge completion

Aug 2026

EF SET English Certificate · C1 Advanced 65/100 · C2 Listening · English in daily professional use at Banco Inter

May 2026

Lean Six Sigma Yellow Belt Thoth / CSSC

2026